La Scuola d'Italia / Dr. Huson / IB Math 12
4 November 2026

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3.3 Pre-Test: Trigonometric Functions

A calculator is allowed. Give answers correct to three significant figures, unless they are exact. Angles are in radians unless they are given in degrees. Diagrams are not to scale.

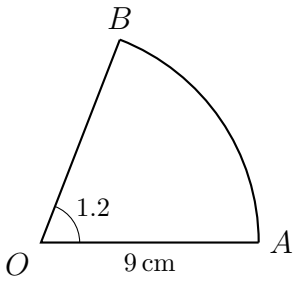
1. [Maximum mark: 9]

arc length $l = r\theta$	area of a sector $A = \frac{1}{2}r^2\theta$

(a) Write 135° in radians, as an exact multiple of π. [1]

(b) Convert 2.4 radians to degrees. [2]

The diagram shows a sector AOB of a circle with centre O and radius 9 cm. The angle $A\hat{O}B = 1.2$ radians.



(c) Find the length of the arc AB. [2]

(d) Find the area of the sector AOB. [2]

(e) A second sector of the same circle has an area of 60 cm^2. Find the angle of this sector, in radians. [2]

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2. [Maximum mark: 10]

(a) Write down the exact value of

(i) $\cos\frac{3\pi}{4}$; [1]

(ii) $\sin\frac{5\pi}{3}$; [1]

(iii) $\tan\frac{4\pi}{3}$. [1]

(b) Given that $\cos\theta = -\frac{5}{13}$ and $\pi < \theta < \frac{3\pi}{2}$, find the exact value of

(i) $\sin\theta$; [3]

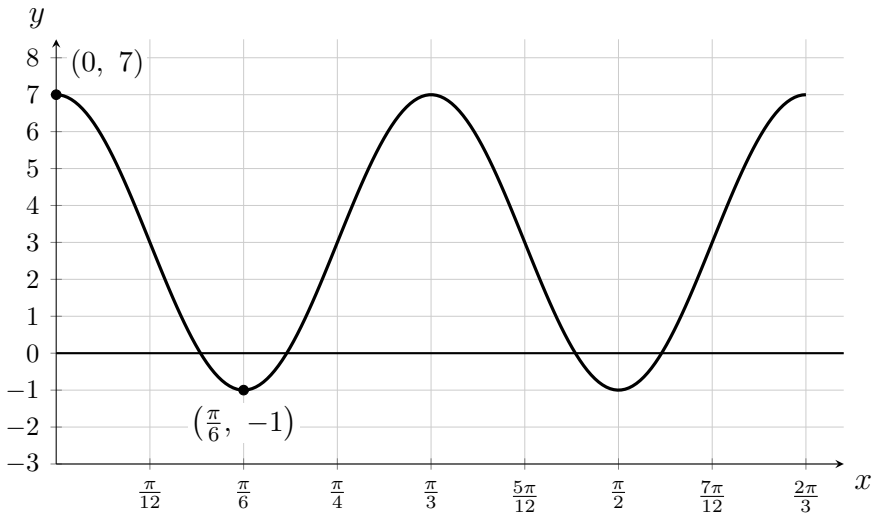
(ii) $\tan\theta$. [1]

(c) Solve the equation $2\sin x + 1 = 0$ for $0 \leq x \leq 2\pi$. Give your answers as exact multiples of π. [3]

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3. [Maximum mark: 10]

The diagram shows part of the graph of $f(x) = a\cos(bx) + d$, for $0 \leq x \leq \frac{2\pi}{3}$, where $a,\ b,\ d > 0$. The graph has a maximum point at $(0,\ 7)$ and a minimum point at $\left(\frac{\pi}{6},\ -1\right)$.



(a) Write down the period of f. [1]

(b) Find the value of a, of b and of d. [3]

(c) The graph of f is stretched vertically with scale factor $\frac{1}{2}$, and then translated 5 units down, to give the graph of h. Find $h(x)$ in the form $h(x) = p\cos(qx) + r$. [3]

(d) Describe fully the sequence of transformations that maps the graph of $y = \cos x$ onto the graph of $y = f(x)$. [3]

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4. [Maximum mark: 11]

The depth of water at the entrance to a harbour, D metres, is modelled by

$$D(t) = 2.5\sin(0.5t) + 6, \qquad t \geq 0,$$

where t is the time in hours after midnight.

(a) Write down the maximum depth and the minimum depth of the water. [2]

(b) Find the period of D. [2]

(c) Find the depth of the water at 09:00. [2]

(d) A ship can enter the harbour only when the depth of the water is at least 7 metres. Find the length of time between $t = 0$ and $t = 12$ during which the ship can enter the harbour. [3]

(e) Find the first time after midnight at which the water is at its minimum depth. [2]